

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2244

Snettisham 138 kV
July 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2244. The Interconnection Customer proposes a 263.3 MW solar photovoltaic facility interconnecting at the Snettisham 138 kV substation in the New Jersey area. The request was received on December 03, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 549 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,854,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Snettisham – Anadale 138.0 kV
Contingency: Snettisham – Fossil Crk 138.0 kV
Loading: 120.3% (Rating: 205 MVA)
DFAX: 100.0%
Pre-project loading: 7.9%
2. Facility: Snettisham Substation – Fossil Crk 138.0 kV
Contingency: Snettisham – Annadale Substation 138.0 kV
Loading: 112.4% (Rating: 220 MVA)
DFAX: 100.0%
Pre-project loading: 7.4%
3. Facility: Ceip – Annadale 138.0 kV L
Contingency: Snettisham – Fossil Creek 138.0 kV
Loading: 104.7% (Rating: 183 MVA)
DFAX: 73.6%
Pre-project loading: 1.5%
4. Facility: Floresville – Fossil Crk 138.0 kV Line
Contingency: Snettisham – Annadale 138.0 kV
Loading: 102.8% (Rating: 235 MVA)
DFAX: 100.0%
Pre-project loading: 9.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Snettisham – Anadale 138.0 kV	Rebuild line with higher-capacity conductor	\$18,582,000	100% (\$18,582,000)
Snettisham Substation – Fossil Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$3,720,000	100% (\$3,720,000)
Annadae – Ceip 138.0 kV Line	Reconductor line	\$2,557,000	86% (\$2,185,000)
Fossil Crk – Floresville 138.0 kV	Reconductor line	\$1,367,000	100% (\$1,367,000)
	Total allocated to project		\$25,854,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.